

Lesson 2D: Solving Quadratic-Form Equations

Criteria	Exceeds Standard (4)	Meets Standard (3)	Approaching Standard (2)	Below Standard (1)
Identification & Substitution	Correctly identifies the quadratic form and chooses the optimal substitution ($u = \dots$) to transform the equation.	Identifies the quadratic form and successfully substitutes to transform the equation, with minor algebraic errors.	Identifies quadratic form but chooses an incorrect substitution or makes significant errors in transformation.	Cannot identify the equation as quadratic form or cannot perform substitution.
Solving the Quadratic	Correctly solves the substituted equation $au^2 + bu + c = 0$ using factoring, quadratic formula, or completing the square.	Solves the quadratic equation with minor calculation errors in factoring or formula application.	Attempts to solve the quadratic but makes multiple errors, or chooses an inefficient method.	Incorrectly solves the quadratic equation or provides no solution steps.
Back-Substitution & Solution	Accurately substitutes u back to the original variable x , solving all resulting equations completely.	Substitutes back to original variable but makes minor errors in final steps or misses one solution.	Substitutes back but does not solve for x , or fails to realize the need to substitute back.	Does not substitute back to the original variable.

Verification & Domain	Identifies extraneous solutions, confirms valid solutions, and checks for domain restrictions.	Identifies most solutions, but may include one extraneous solution or miss one.	Fails to recognize extraneous solutions; fails to check solutions.	Does not verify solutions.
Communication & Method	Work is logical, organized, and clearly justifies each step using proper mathematical notation.	Work is mostly organized and understandable, with minor, inconsequential errors in notation.	Work is disorganized, making it difficult to follow the logical flow of solving.	Work is missing, or steps are illogical.